```
-- Wed/Thu/Fri --> 9 AM onwards
-- sat/sun --> 8:30 AM onwards



For every warehouse, calculate:


WarehouseID -->warehouse
Warehouse name --> warehouse
Number of orders handled --> orders
Total quantity of products sold --> order_items
Total sales value --> order_items


-- importance -->warehouse table
Include warehouses even if they have no orders.


Display only warehouses where the total sales value is greater
than ₹2,00,000. -->having


Sort from highest to lowest total sales. --> order by

select *
from `e1.warehouses`
limit 5
```

```sql
alter table `e1.warehouses`
rename column string_field_0 to WarehouseID

alter table `e1.warehouses`
rename column string_field_1 to Warehouse




select
  w.WarehouseID,
  w.Warehouse,
  count(distinct o.OrderID) as no_of_orders,
  coalesce(sum(oi.Total),0) as total_sales,
  coalesce(sum(oi.Quantity)) as total_qty
from `e1.warehouses` as w
left join `e1.orders` as o
  on w.warehouseID = o.WarehouseID
left join `e1.order_items` as oi
  on o.OrderID = oi.OrderID
group by
  w.WarehouseID,
  w.Warehouse
having total_sales > 200000
order by total_sales desc
```

```
Calculate the total sales value generated by every customer.

Then display only those customers whose total spending is
greater than the average customer spending.

Display:

CustomerID --> customer
Customer name --> customer
Total spending --> order_items

Sort from highest spending to lowest.



average customer spending.

c1 --> s1
c2 --> s2
c3 --> s3


-- avg cust spending --> (s1+s2+s3)/3
why not avg(total)


select
  c.CustomerID,
  c.Name,
  sum(oi.Total) as total_sales
```

```sql
from `e1.customers` as c
inner join `e1.orders` as o
  on c.CustomerID = o.CustomerID
inner join `e1.order_items` as oi
  on o.OrderID = oi.OrderID
group by
  c.CustomerID,
  c.Name
having total_sales > (
select
  round(avg(total_sales),2)
from  (select
          o.CustomerID,
          sum(oi.Total) as total_sales
        from `e1.order_items` as oi
        inner join `e1.orders` as o
          on oi.OrderID = o.OrderID
        group by o.CustomerID
      )
    )
order by total_sales desc
```

For every product, calculate its total sales value.

Display only products whose total sales value is greater than the average total sales value of products belonging to the same category.

```
Display:

ProductID --> product
ProductName --> product
CategoryName --> category
TotalSales --> order_items

Sort by total sales from highest to lowest.

p1 -> s1
p2 -> s2
p3 -> s3

select
  p.ProductID,
  p.ProductName,
  c.CategoryName,
  sum(oi.Total) as total_sales
from `e1.products` as p
inner join `e1.categories` as c
  on p.CategoryID = c.CategoryID
inner join `e1.order_items` as oi
  on p.ProductID = oi.ProductID
group by
  p.ProductID,
  p.ProductName,
  p.CategoryID,
  c.CategoryName
having total_sales > (select
```

```
  avg(total_sales) as avg_sales
  from (
        select
          p1.ProductID,
          sum(oi.Total) as total_sales
        from `e1.products` as p1
        inner join `e1.order_items` as oi
          on p1.ProductID = oi.ProductID
        where p1.CategoryID = p.CategoryID
        group by p1.ProductID

  ))
order by total_sales desc
```

For every order, calculate the total order value by adding the Total column from order_items.

Then classify each order as:

High Value → Total order value ≥ ₹50,000
Medium Value → Total order value between ₹20,000 and ₹49,999
Low Value → Total order value < ₹20,000

Display:

OrderID

```
OrderDate
CustomerName
TotalOrderValue
OrderCategory

Show the highest-value orders first.

-- case when statements
-- if-elseif-else

syntax:

case
  when condition1 then result1
  when condition2 then result2
  when condition3 then result3
  else default_result
end



select *
from `e1.customers`
limit 5



categorise customers as
18-20: genz,
21-40 : adult,
```

```
41 onwards : senior_citizen

select
  CustomerID,
  Age,
  Name,
  case
    when Age >= 18 and Age <= 20 then "GenZ"
    when Age >= 21 and Age <= 40 then "Adult"
    else "Senior_Citizen"
  end as age_group
from `e1.customers` as c



select
  CustomerID,
  Age,
  Name,
  case
    when Age >= 18 and Age <= 20 then "GenZ"
    when Age <= 40 then "Adult"
    when Age >40 then "Senior_Citizen"
  end as age_group
from `e1.customers` as c


select min(age) from `e1.customers`

select *
```

```
from `e1.customers`
limit 5

F --> Female
M --> Male




This is the hardest integrated question.

Problem Statement

Create a category-level sales report containing:

CategoryID
CategoryName
Number of products in the category
Number of different products sold
Total quantity sold
Total sales value
Average selling price

Only consider order-item records where:

SellingPrice > ₹10,000

Display only categories where:

Total sales value > ₹5,00,000
```

```
Sort categories from highest to lowest total sales.
```